



FIELD HOW TO

MX60 Mobile Mapping



Trimble MX60 · Trimble Mobile Imaging

DOCUMENT

MX60 Field How To

SYSTEM

Trimble MX60 · Trimble Mobile Imaging

SET

Technical Manual · Standard Operating Procedure · Field How To · Office How To

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FIELD HOW TO

MX60 Mobile Mapping

How to run the MX60 in the field. Short numbered steps, meant to be used in or near the vehicle.

TRIMBLE MX60 · TRIMBLE MOBILE IMAGING

Start with Appendix E, the Quick Card — the ten things that cost the most if missed. Four of them cannot be fixed from the office: an aiding sensor that was never activated, missed overlap, a mission with no closing sequence, and a disk cleared before the copy was verified.

CAUTION Data loss, an irreversible operation, a lost result, or safety	IMPORTANT Gets the job wrong if ignored	IN PLAIN LANGUAGE What the section means, why it matters, what to remember
TRIMBLE METHOD Trimble documents this — NOT NECESSARILY A REQUIREMENT	RECOMMENDED PRACTICE Parametrix recommendation — FOLLOW IT UNLESS THE PROJECT SAYS OTHERWISE	SET BY THE PROJECT No single company-wide answer. The project decides, and records it
NO PUBLISHED FIGURE No published number exists. Judgement, and the evidence on the job		

1 How to Use This Guide

1.1 Shape

Short numbered steps for a survey employee who may be using the MX60 for the first time under the appropriate supervision. A **Why this matters** line appears only where the reason changes what you do. The guide teaches the machine and the workflow, not surveying. Deeper theory stays in the Technical Manual.

Every subsection is numbered. When reporting a problem with this guide, cite the number — **Field How To §12.3** — together with any **W-**, **D-**, **T** or **V-** identifier on the statement you mean. Front matter explains why.

1.2 Order

§2 to §11 get the system on the vehicle and running. §12 to §14 initialize it. §15 to §20 collect. §21 to §26 finish, check and hand over. §27 is what to check when something is wrong.

1.3 The five that cost the most

All five are silent, and four of them cannot be fixed from the office.

		§
1	An aiding sensor not activated in Vehicle Settings logs nothing — even if it is installed and wired	10
2	Green is not finished. Allow the settling time before recording anything that matters	14
3	The closing sequence cannot be added later	21
4	Missed overlap removes office remedies — and nothing warns you	17
5	Do not clear a disk until the office confirms a verified copy	25

Full card: **Appendix E.**

1.4 If something is not covered here

Stop and ask. Do not improvise on the vehicle and do not turn an MX60 problem into a guess. Use §27 first: identify the symptom, restore the last known good state, check one thing at a time, and record what was abnormal.

Three things you may always do without asking first:

Stand down on safety or data-quality grounds[SOP §9.5]	The decision and its reason are recorded. This is not conditional on you being right. Nobody has to wait for anything to stop an unsafe job
Record a comment about anything unusual	§19
Re-drive a run while you are still on site	§24

1.5 Who is telling you — the authority key

This guide cannot require anything. Everything in it is somebody else's instruction, and the marker says whose. That matters because the two are not the same: a Parametrix practice is a recommendation, while **a Trimble instruction and an equipment limit bind regardless.**

MARKER	WHO SAYS SO	FORCE TODAY
[TRIMBLE]	Trimble states it as a requirement, in the cited manual page or topic	Binding. Does not wait on a Parametrix decision
[TRIMBLE METHOD]	Trimble documents the method, but does not state it as a requirement	Strong advice. The method is Trimble's; the obligation is not
[EQUIPMENT]	A hardware or safety limit, with manufacturer evidence	Binding. It is a fact about the machine
[SOP §n]	A Parametrix requirement, at that clause	As strong as that clause — check its state
[RECOMMENDED]	Recommended by this project	Not company policy. Do it unless told otherwise, and say so if you cannot
[TESTING · Tn]	Depends on a result nobody has yet	An interim posture, not a rule
[DECISION · D-n]	Parametrix has not decided	Ask. Do not improvise a standing rule

[TRIMBLE] and [TRIMBLE METHOD] are not the same thing. Trimble writes "*Navigation alignment must be done first*" — that is **[TRIMBLE]**. It heads its in-field list "*Proposal of a checklist for system operation*" — that is **[TRIMBLE METHOD]**. This guide does not promote the second into the first. A documented method is the best method anyone has published, and it is still not a manufacturer requirement.

1.6 Where a decision is open

A **SET BY THE PROJECT** marker means the SOP has identified a requirement whose answer is not set. Where this guide suggests what to do meanwhile, it is a suggestion and not a Parametrix standard.

1.7 The field record

No software produces it. Conditions, incidents, what was not collected and why — if you do not write it down, nothing else will. §26 and Appendix C.

1.8 First supervised mission

For a person with minimal MX60 experience, use the guide in this order:

1. Read the **Quick Card** in Appendix E and complete the **Preflight Checklist** in Appendix A.
2. Work through §§2–11 in order. Do not skip ahead because the system happens to power on.
3. Initialize with §§12–14 and confirm that the status you see matches the guide before recording.
4. Collect with §§15–20 open and use §27 immediately if an indication is not normal.
5. Perform the closing sequence and shutdown with §§21–23.
6. Make the re-drive decision while still on site using §24.
7. Transfer and hand off with §§25–26. The job is not finished when the vehicle is parked.

The goal is not memorization. The goal is that the operator can tell normal from abnormal, follow the recovery path, and identify the point where continuing would risk the data.

2 Before You Leave the Yard

1. **Read the mission plan.** Route, passes and their directions, planned overlap, GNSS-hostile stretches, and the two initialization locations (*SOP §8*)
2. Confirm the **primary and backup initialization locations** are scouted, not assumed
3. Confirm the **data disk** is the intended one and has space (§11)
4. Confirm **calibration currency** — when was the system last calibrated? (*SOP §15.2*)
5. Take the **field record form** (Appendix C) and this guide
6. Confirm the vehicle's power supply is sound (§6)

WHY THIS MATTERS — THE TWO INITIALIZATION LOCATIONS

Discovering the chosen lot is fenced, occupied or under trees costs twenty minutes at the worst moment of the day. Scout two on imagery before mobilising (*Technical Manual §15.4*).

SET BY THE PROJECT

Base station strategy — whether a local base must be occupied for this mission depends on whether Parametrix works in **IN-Fusion+ Single Base** or **PP-RTX**, which is not decided (*SOP §6.2, §8.5*). **Check the mission plan for this job.**

Stop if

- There is no mission plan
- The calibration date cannot be established
- The segments mobile mapping is **not** expected to serve have not been identified

(*SOP §8.6*)

3 Equipment Inspection

3.1 Look at

ITEM	LOOKING FOR
Optical surfaces — scanner windows, camera dome	Rain, dust, insects, smears, scratches. They export as defects you cannot fix
Cables	Chafe, cuts, bent pins, strain at the connector
Mounting rack and fixings	Anything loose. Check against the tightening method in the rack's own user guide
Sensor Unit seating	That it sits as it did when the lever arms were measured
GAMS antenna , if fitted	Secure, and the same antenna type as the primary MX60 UG Rev B, p.68
DMI , if fitted	Secure on a non-steering wheel
Data disk	Present, correct, seated

3.2 Clean the optics before every mission

A smear on the dome is in every image of the day. It is thirty seconds now.

3.3 Stop if

- An optical surface cannot be cleaned properly
- A cable is damaged
- **Anything about the Sensor Unit's seating is in doubt.** A doubt about seating is a doubt about the lever arms, and a lever arm error is systematic and invisible (*Technical Manual §7.6*)

The Control Unit and Power Unit are IP30 — not waterproof

MX60 UG Rev B, p.53. They live inside the vehicle.

4 Mounting the Sensor Unit

CAUTION

Two people are required. The Sensor Unit weighs **24–28 kg** depending on configuration (*MX60 UG Rev B*). It is awkward, it is expensive, and it is lifted above head height.

4.1 Do

1. Two people. Plan the lift before starting it
2. Seat the unit on the rack in the orientation it was installed and measured in
3. Secure it per the rack's documented tightening method and sequence
4. Connect the Sensor Unit cable — **5 m** (*MX60 UG Rev B*)
5. Confirm the unit is seated exactly as it was when the lever arms were measured

4.2 Why this matters

The lever arms — the fixed distances between sensors — are **measured, not computed**. The calibration solves angles only, so a seating change cannot be calibrated out later. It becomes a systematic error nobody will attribute to its real cause (*Technical Manual §7.6*).

CAUTION

Changing the rack, the roof bars, the vehicle, or the Sensor Unit's position on the rack invalidates the lever arms and may invalidate the calibration. If any of those has changed, say so before the mission rather than after it.

4.3 Stop if

- You are alone
- The seating is not repeatable, or the rack fixings are not as documented

SET BY THE PROJECT

Does removing and refitting the Sensor Unit count as disturbing the calibration? If the unit comes off between jobs, the answer decides whether calibration is annual or per-mobilisation (*SOP §15.2*).

SET BY THE PROJECT

Where are the lever arms, the Vehicle Preset and the installation configuration recorded, and who verifies them? (*SOP §9.1*)

5 Connections and Cable Routing

5.1 Do

1. Route cables so they cannot **chafe, catch, or be closed in a door**
2. Secure them — a cable that moves will eventually fail at the connector
3. Sensor Unit run: **5 m**. Control Unit run: about **3 m** (*MX60 UG Rev B*)
4. Confirm every connector is fully seated
5. Keep the Control Unit and Power Unit **inside the vehicle** — they are IP30, not waterproof

5.2 Stop if

CAUTION

Never connect the USB cable while the exchangeable data disk is inside the Control Unit. Remove the disk first *MX60 UG Rev B*, p.10.

Also stop if a connector will not seat fully, or a cable run cannot be secured away from a door seal or a moving part.

6 Power System Checks

6.1 The numbers

	VALUE	SOURCE
Input voltage	12–16 V DC	MX60 UG Rev B
Current at startup	25 A at 12.8 V (320 W)	MX60 UG Rev B
Current in operation	12 A (160 W)	MX60 UG Rev B
Supply rating needed	30 A or more	MX60 QSG Rev B, p.4
Direct-connection fuse	35 A , close to the battery	MX60 UG Rev B

6.2 Do

1. Confirm the supply is rated and fused as above
2. Confirm the battery is in good condition and charged
3. **Keep the engine running while the system is operating**
4. Confirm the auxiliary battery, if fitted, is healthy

An auxiliary battery as a backup power source is recommended

MX60 QSG Rev B, p.4.

6.3 Why this matters

A supply interruption mid-mission does not merely stop collection. **It ends the run, and takes the closing sequence with it** — which is the one thing that cannot be added afterwards (§21).

6.4 Stop if

- The battery is marginal. **A mission that starts on a marginal battery will end unexpectedly**
- The supply is not rated for 30 A, or is not properly fused

7 Battery Protect

CAUTION

Battery Protect MX60 UG Rev B, p.27:

Event | Trigger

---|---

Audible warning | Supply below **10.5 V for longer than 12 seconds**

Power cut | Supply below **10.5 V for more than 90 seconds**

Recovery | Voltage rises above **12.0 V within that 90 seconds**

The alarm leaves roughly **78 seconds** to restore charge — that figure is the arithmetic of the two sourced timings, not a separately published one.

An interrupted run loses the closing sequence with it.

7.1 If the alarm sounds

1. **Restore charge immediately.** Engine speed up, load off
2. Do not continue collecting while you work out what happened
3. Record it in the field record — time, what you were doing, what you did

7.2 If it sounds during preflight

Fix it before doing anything else. Do not start a mission on a battery that has already tripped the warning.

7.3 If power cut mid-run

The run is ended	It has no closing sequence
The mission may be incomplete	Check what was written (§23)
This is a re-drive decision, on site	§24

7.4 Treat the alarm as an instruction

Not as information. You have about a minute and a quarter.

8 Starting the System

8.1 Do

1. **Vehicle ignition on**
2. Confirm the power supply is live
3. Press and **hold the Control Unit power button for at least 15 seconds**
4. Sensor Unit and Control Unit LEDs **blink for about 10 seconds**
5. **Wait for the system to reach a ready state** before connecting to TMI

(MX60 UG Rev B; MX60 QSG Rev B)

8.2 Expect

Blinking LEDs settling to a steady state. Give it time — a firmware update can take **up to six minutes** to complete on power-up TMI UG Rev L, p.49.

8.3 Stop if

- The LEDs do not settle
- The Battery Protect alarm sounds (§7)
- The system does not reach a ready state

Do not remove power or the data disk while the power button light is on (§22).

9 Connecting to TMI

9.1 Do

1. Open **Chrome**
2. Go to `http://tmi.mx-scan.net` — the capture interface
3. Administration, if needed: `http://admin.mx-scan.net`

(TMI UG Rev L)

TMI is served by the Control Unit. It needs no internet access, and the addresses resolve only on the Control Unit's network.

9.2 The status colours

Learn these before driving. This interface is your only view of system health.

COLOUR	MEANING
Green	The parameter meets its accuracy threshold
Orange	Degraded but operating. Recording is permitted
Red	Not meeting threshold

CAUTION

Orange permits recording. Survey-grade work does not.

TMI will let a mission be recorded on an orange navigation status. Whether Parametrix work may be collected on anything other than green is / — an operator decision rule that has not been made. **Until it is, treat orange as a stop-and-assess condition and record it.**

9.3 Confirm every sensor is present

Check the device list. Every camera and both lasers.

A sensor absent from the list is a cable, power or sensor fault. A run collected with a sensor down is incomplete, and you will not know which part of the deliverable is missing until the office opens it.

9.4 If TMI will not load

Check these in order:

1. Confirm the **Control Unit is powered and has completed startup** (§8)
2. Confirm the field device is connected to the **Control Unit's network** — TMI does not require internet access
3. Enter the capture address exactly: `http://tmi.mx-scan.net`
4. Use **Chrome**
5. If the connection is still not stable, return to the documented startup/connection sequence rather than changing network or system settings at random

Stop if TMI cannot be reached reliably or the expected device list cannot be confirmed.

CONFIGURATION NOTE

The exact connection method, network identifier and complete expected device list must be confirmed against the actual Parametrix unit before independent use. Do not invent an SSID, IP address or sensor list that has not been observed.

Full status reference: **Appendix D**.

10 Mission Setup in TMI

10.1 Vehicle settings

CAUTION — • [TRIMBLE]

If an aiding navigation sensor is not activated in Vehicle Settings, its data will NOT be logged — even though all connections may have been made properly TMI UG Rev L, p.21.

This is Trimble stating how the system behaves. It is not a rule anybody can relax.

This applies to **DMI and GAMS**. The hardware can be correctly installed, wired and present, and log nothing. There is no cabling fault to find and nothing looks wrong.

1. Open **Vehicle Settings**
2. **Confirm DMI is activated**, if fitted — and that its lever arm and **mounting position, left or right**, are set
3. **Confirm GAMS is activated**, if fitted — and that its lever arm is set
4. Confirm the **Install Height** preset is the one for this vehicle

The mounting side matters in the office. The DMI's scale factor sign depends on it, and the processor cannot see the vehicle. **Record which side it is on** (*Technical Manual §9.2*).

10.2 Capture settings

CURRENT CONFIGURATION NOTE

The installed TMI build on the Parametrix unit has not yet been confirmed. The two manufacturer-documented presentations below are retained until the actual unit is commissioned. Before independent use, this section should be reduced to the presentation employees actually see. During supervised setup, record which presentation is present.

Source | What the operator sees

---|---

MX60 Quick Start Guide Rev B, p.10 | Two controls — **Measurement Prog**

[500 kHz, 1000 kHz] and **Line Speed** [120 Hz, 200 Hz]

TMI User Guide Rev L, p.29 | A single combined **Laser Mode**

Which you get depends on the TMI version installed, which has not been confirmed (, CONFLICT-005).

SETTING	RANGE	NOTE
Laser rate	500 / 1000 kHz per scanner	The higher rate costs range: 120 m instead of 150 m
Line speed	120 / 200 Hz per scanner	More profiles per second = closer spacing along the corridor
Lateral Range Limit	5–50 m	— whether it affects accuracy or is purely a data-volume control is not documented
Dust filter	on/off	For unpaved roads and mine sites (<i>Product Bulletin, January 2025</i>)

The measurements-per-second values TMI displays are **rounded**

TMI UG Rev L, pp.28–29.

The specification sheet's 1000/2000 kHz and 240/400 Hz are system totals across both scanners. TMI uses the per-scanner numbering above. Do not go looking for 2000 kHz on the screen.

NO PUBLISHED FIGURE

The DMI scale factor's 5 % default assumes somebody measured the wheel. If the value came from a manual for a nominal tyre, the office is weighting the DMI on an unverified claim. **If you know whether this wheel was measured, record it** (*SOP §13.1*).

10.3 Record

Capture settings used, **and which presentation you saw**. Field record, Appendix C.

10.4 Stop if

- A fitted aiding sensor cannot be activated
- You cannot establish which capture settings are in force

11 Disk Check

11.1 Do

1. Confirm the exchangeable data disk is **installed and seated**
2. Confirm it is **the intended disk**
3. Confirm **free space** is sufficient for the planned mission, with margin

11.2 Stop if

CAUTION

Never connect the USB cable while the exchangeable data disk is inside the Control Unit. Remove the disk first MX60 UG Rev B, p.10.

Also stop if free space is marginal.

WHY THIS MATTERS

A mission that fills the disk mid-corridor ends the run and takes the closing sequence with it (§21). Storage is 2 × 4 TB removable SSD (*MX60 Spec Sheet*); running out is a planning failure, not an accident.

SET BY THE PROJECT

What free-space margin is required before a mission may start? (SOP §9.4)

11.3 Never reuse a disk on a promise

Do not clear a disk because someone said the data was copied. §25 has the rule: not until the office confirms a verified copy exists in two places and the `.mxd` opens.

12 Navigation Initialization

The sequence, in order. Do not compress it.

12.1 Do

1. **Park in an open-sky area** with good GNSS visibility, away from high buildings and canopy
2. **Start the mission** in TMI
3. **Log 2–3 minutes of static data** before driving
4. **Drive straight ahead for approximately 20 m**, with no larger steering. The navigation status switches on completing this
5. **Drive straight at varying speed — accelerate then decelerate — and perform dynamic steering manoeuvres.** An example profile: **0 → 50 → 20 → 50 → 20 km/h**
6. Watch the navigation status progress **red → orange → green**
7. **Allow up to 10 further minutes of settling before logging data that matters**[TRIMBLE METHOD] MX60 QSG Rev B, §5.3 p.11; §6 p.14 — steps 1–7 are Trimble's documented

sequence. Trimble writes "*should be started in a static mode*" and heads its checklist "*Proposal of a checklist for system operation*", so the sequence is documented method, not a manufacturer requirement. **Do it anyway** — nobody has published a better one, and the office cannot repair a bad initialization.

[TRIMBLE] MX60 QSG Rev B, §5.3 p.11 — one part of it is a requirement and the system enforces it: "*Navigation alignment must be done first before data logging is allowed!*" You cannot log through step 6 whether you want to or not.

Step 7 is the one nothing enforces (§14.2), and it is advice in Trimble's own words — "*it is advised to add some time (up to 10 minutes)*". It is also where most of the value is.

12.2 What each step is doing

One line each. Full explanation: **Technical Manual §13.**

STEP	DOING
Static period	Parked, the system's true speed is zero, so anything the motion sensors report is pure error it can measure and remove
Straight 20 m	Initial heading from the direction of travel. This is what the status is waiting on

STEP	DOING
Varying speed	Separates an accelerometer bias from a pitch error
Dynamic steering	Heading — the hardest component, and what the turns are for

12.3 Stop if

- **Open sky is not available at the start point.** Drive to open sky *before starting the mission*, not after
- The status will not progress. See §14 and Appendix D — ask TMI **which** parameter is holding it

12.4 The system enforces one part of this, and not the other

Navigation alignment must complete before data logging is allowed — the system enforces that (*MX60 QSG Rev B*).

It does not enforce step 7. That one is yours (§14).

IN PLAIN LANGUAGE

What this section means. The start-of-mission routine is not the system warming up. It is the system solving for things it cannot measure any other way, using your driving as the input.

Why it matters. Parked, it knows its true speed is zero, so everything the motion sensors report is error it can measure and remove. Driving straight separates one kind of error; changing speed separates another; turning is the only thing that fixes heading, which is the hardest of the three and the one that hurts most at range.

Remember this. Do the whole profile, in order, and then wait. Green means the solution met its accuracy thresholds — not that it has settled.

If this is skipped. The first part of the job is collected on the weakest solution of the day. Nothing warns you, and no office step recovers it.

13 GAMS Considerations

Only if GAMS is fitted. Whether it is on this system is an open question (*SOP §6.4*).

13.1 What it changes

TRIMBLE METHOD — MX60 UG REV B, P.67

GAMS reduces initialization time and eliminates the special driving manoeuvres otherwise required.

Read the second half. By implication, the manoeuvres are required when GAMS is not fitted. Trimble says the same from the other direction: **straight driving is more important if a GAMS antenna is not used**

MX60 QSG Rev B, p.12.

13.2 What to do

RECOMMENDED PRACTICE — DELIBERATELY CONSERVATIVE UNTIL FITMENT IS CONFIRMED

Perform the full sequence either way (§12). Trimble documents that GAMS eliminates the special driving manoeuvres; this guide is not claiming otherwise. The extra manoeuvres are retained because they cost little, the static period still matters, and the delivered GAMS fitment has not yet been confirmed.

After the actual system configuration and first controlled missions are validated, Parametrix can either retain this conservative practice or simplify the sequence to match the confirmed configuration.

13.3 If GAMS is fitted — the installation requirements

	VALUE
Offset accuracy, collection only	10 cm or better
Offset accuracy, post-processing	A few millimetres
Minimum baseline	2.0 m between primary and secondary antennas
Antenna matching	Must be the same type as the primary. Do not mix
Measured to	The L1 antenna phase centre of the secondary antenna

VALUE	
Re-measure	Every time GAMS is re-installed on the roof for a new mission

MX60 UG Rev B, pp.67–68

All Parametrix mobile mapping is post-processed, so the requirement is millimetres, not centimetres. An offset good enough to navigate with is not good enough to survey with.

13.4 Stop if

- GAMS is fitted but not activated in Vehicle Settings (§10.1) — **it will log nothing**
- The published rack corner offsets are being used and you are not certain which rack is fitted. Those offsets apply to the **standard Trimble Roof Rack only**

MX60 UG Rev B, p.68

NOT DOCUMENTED BY TRIMBLE

The **Trimble GAMS Antenna Kit Installation & Operation Manual** is not held. It is needed to complete the installation procedure if GAMS is fitted.

14 Reading Navigation Status

14.1 The colours

COLOUR	MEANING
Red	Not meeting threshold
Orange	Degraded but operating. Recording is permitted
Green	The parameter meets its accuracy threshold

14.2 What green does not mean

IMPORTANT — · [TRIMBLE]

Green means the solution met the accuracy thresholds — not that it has finished converging. That is why Trimble asks for **up to ten more minutes** before recording anything that matters (*MX60 QSG Rev B*).

The first data after the light turns green is the weakest data of the day. Do not spend it on the most important part of the corridor.

The light does not change again when the estimate improves, so nothing tells you. This is the one part of initialization the system does not enforce, and therefore the part that gets skipped.

14.3 If it will not reach green

Ask TMI which parameter is holding it. The remedies are different and waiting helps neither.

HOLDING	LIKELY CAUSE	DO
Heading	Not enough dynamic manoeuvres, or no GAMS	More turns and speed changes
Position	Poor sky view	Move the vehicle to genuinely open sky
Attitude	Not enough motion variety	Complete the full manoeuvre profile

14.4 Orange

Orange permits recording. Survey-grade work does not.

Until / is decided, treat orange as a **stop-and-assess** condition: work out why, record it, and decide deliberately rather than by default.

14.5 Record

The time green was reached, and the time recording of significant data began. The gap between them is the settling time, and it is the evidence that §14.2 was respected.

15 Recording Runs

15.1 Do

1. Start a run at the start of a continuous stretch of collection
2. Stop it at the end of that stretch
3. Start the next one
4. Note each run's start and end, and anything that happened during it, in the field record

Each run becomes a **Run** node in the office.

15.2 Minimum mission time

TRIMBLE METHOD

Minimum mission time: 30 minutes MX60 QSG Rev B, pp.13–14.

A short corridor does not excuse a short mission. If the collection itself is twelve minutes, **keep the system running and logging navigation data to reach thirty.** The trajectory solution improves with observation time, and closing early gives the office less to work with.

15.3 Drive every planned pass, in the planned direction

Direction matters. A second pass in the opposite direction fixes the far side of the corridor better than any setting change, because it turns grazing incidence into direct incidence (*Technical Manual §16.3*).

SET BY THE PROJECT

How many passes, in what pattern, by roadway type? The mission plan states it for this job; the standing rule does not exist yet (*SOP §8.2*).

15.4 Stop if

- Navigation status will not hold
- A sensor has stopped reporting
- Storage will not last the corridor
- The battery is cycling into protection

You have the authority to stand down (SOP §9.5). A mission abandoned after twenty minutes costs twenty minutes. A mission completed on a degraded solution costs the office days and may cost a return visit.

16 Driving Practices

16.1 Speed

	VALUE	AUTHORITY
Recommended maximum with the system operating	80 km/h (50 mph)	[TRIMBLE] (MX60 UG Rev B) — Trimble's recommendation, not a Parametrix rule
Absolute maximum, operating or not	110 km/h (68 mph)	[EQUIPMENT] (MX60 UG Rev B)

SET BY THE PROJECT

What collection speed, by deliverable type? Trimble publishes a recommended maximum and an absolute maximum and **no guidance relating speed to deliverable quality** (SOP §9.4).

Recommended, where the project does not say otherwise: collect at or near prevailing traffic speed up to 80 km/h, reducing where point density requires it.

[TRIMBLE METHOD] Trimble recommends not exceeding 80 km/h with the system operating. That recommendation is not waiting on and it stands today — but it is Trimble's *recommended* maximum, not a stated limit, and this guide does not upgrade it into one. **110 km/h is the limit**, and that one is **[EQUIPMENT]**.

16.2 Smoothness

Drive smoothly. Sudden braking and sharp manoeuvres stress the inertial solution unnecessarily.

16.3 Lane choice

Choose the lane with the best line of sight to the features being collected.

The far side of a truck is not collected at all. Occlusion is permanent — no office step recovers it.

16.4 Heat

CAUTION

Direct sun with the vehicle stationary or driving below 10 km/h is outside the rated operating envelope MX60 UG Rev B, p.53.

Extended idling in direct sun — at a signal, in a queue, waiting for traffic control — is a real risk on a hot day.

16.5 Turning round

Plan turnarounds where the vehicle can complete them **without reversing under the sensor's collection**. Where a turn must happen inside the corridor, **note it in the field record** so the office knows why the trajectory does what it does there.

16.6 Record

Anything that changes what was collected: a lane you could not use, a queue you sat in, a turnaround inside the corridor.

17 GNSS While Driving

17.1 What to watch

Navigation status, continuously. Any degradation from its ready state.

17.2 The hostile stretches

Your mission plan identified them (§2). As you reach each one:

1. Note the time entering and leaving — **duration is what matters, not length**
2. Drive it as planned, in the planned direction
3. **Drive the planned overlap**
4. Record the conditions actually encountered

Duration, not length. Inertial drift is a function of time. A 300 m tunnel at 80 km/h is 13 seconds; the same tunnel at 20 km/h in traffic is nearly a minute (*Technical Manual §15.1*).

17.3 The one that removes office options

CAUTION

If a GNSS-hostile stretch planned for two passes receives one, **both LiDAR QC and run-to-run registration become unavailable** for that stretch. There is no software warning, and no way to tell afterwards except by looking at what is there.

If you cannot drive the planned overlap, say so in the field record and say so at handoff. It is not recoverable in the office and it is cheap to fix while you are still here.

NO PUBLISHED FIGURE

Whether predicted GNSS conditions correlate with achieved trajectory RMS on this system. Until that is established, the planning estimate is judgement — **so what you observe and record here is the evidence that will settle it** (*SOP §8.3*).

17.4 The published limit

Trimble publishes positioning performance at **no outage** and after a **60-second outage**, and nothing beyond MX60 UG Rev B, p.56.

An outage materially longer than 60 seconds is a planning problem, not a driving problem. If you meet one that was not planned for, record it and raise it.

17.5 Record

Every hostile stretch: entry and exit time, conditions, whether the planned overlap was driven.

IN PLAIN LANGUAGE

What this section means. The satellites are the only thing holding the trajectory to the ground. Under trees, beside tall buildings and in tunnels, the system is running on its inertial sensors alone and drifting.

Why it matters. Drift is not random noise that averages out. It grows for as long as the sky is blocked, and every point collected during that stretch inherits it. A short gap between good sky is bridged well. A long one is not.

Remember this. Watch where the sky closes in and note it. If a stretch was bad, say so in a Comment as it happens — ten seconds in the vehicle saves an hour of office guesswork later.

If this is skipped. The office sees a degraded stretch with no explanation, and has to decide between re-collecting it and delivering it without knowing why it is poor.

18

Field QC Indicators

18.1 What the field can and cannot confirm

CAN CONFIRM HERE	CANNOT CONFIRM UNTIL THE OFFICE
The mission recorded and closed	Post-processed trajectory quality
All planned runs exist	Accuracy against control
Data written to disk and readable	Whether GNSS degradation was bridged adequately
Coverage — was the corridor driven	Point density at the achieved standoff
Sensors reported present throughout	Imagery detail sufficiency
Gross imagery problems — obstruction, contamination	Colour fringing, boresight symptoms
Free space and disk health	Anything requiring the trajectory

IMPORTANT

Nothing available in the field confirms the data is of survey quality. The field checks confirm that the data **exists, is complete, and is readable**. That is genuinely worth confirming before leaving — and it is not the same thing.

18.2 Watch, while driving

WATCH	FOR
Navigation status	Any degradation from the ready state
Storage	Remaining capacity against remaining corridor
Sensor status	A camera or laser that has stopped
Audible alarm	Battery Protect (§7)

18.3 Recognising a bad collection before the day is wasted

Stop and reassess if any of these is true:

- Navigation status will not hold
- A sensor has stopped reporting
- Storage will not last the corridor
- The battery is cycling into protection
- Conditions have moved outside the go/no-go rule

The operator has authority to stand down (*SOP §9.5*).

19 Using Comments

19.1 Do

Use TMI's Comments feature as things happen. Not afterwards.

Examples worth recording:

- *"heavy canopy from the bridge"*
- *"stopped 4 min, traffic control"*
- *"parked truck occluding the north side"*
- *"could not drive second pass, road closed"*
- *"hard rain from here"*

19.2 Why the office needs them

Nothing in the software knows any of this. TBC can see the trajectory degraded; it cannot see that a truck was there. The office will otherwise spend an hour inferring from residuals what you could have written in ten seconds (*Technical Manual §30*).

19.3 The test

If you find yourself thinking *"the office will wonder why the data looks like that here"* — **that is a comment.**

19.4 Comments are not the field record

Both are needed. Comments are timestamped notes inside the mission; the field record is the document that travels with the data (§26, Appendix C).

20 Stopping and Restarting

20.1 A planned break

- 1. Stop the run
- 2. **Leave the mission open and the system logging** if you are returning shortly
- 3. Restart a new run when you resume
- 4. Record the break — start, end, reason

Keeping the mission open keeps the navigation solution continuous.
Closing and reopening starts a new trajectory, and the new one has to be initialized again.

20.2 An unplanned stop

CAUSE	DO
Traffic, obstruction, waiting	Leave the mission open. Record it as a comment (§19)
Battery Protect alarm	§7 — restore charge immediately
Power cut	The run is ended and has no closing sequence. Go to §24
Sensor dropped out	Stop. A run with a sensor down is incomplete

20.3 Heat

If you are stationary in direct sun, remember §16.4 — that is outside the rated operating envelope. Move, or shut down.

20.4 Restarting after a mission was closed

A new mission is a new trajectory. It needs its own initialization (§12) and its own closing sequence (§21). You cannot append to a closed mission.

21 The Closing Sequence

CAUTION

The post-processed trajectory is computed **forward and backward** and merged. A degraded stretch mid-mission is bracketed by good data on both sides. **A degraded stretch at the end has good data on one side only.**

The closing sequence is the reverse pass's anchor. It cannot be added later.

21.1 Do

1. Finish the last run
2. **Drive to an open-sky location — with the mission still running**
3. **Dynamic steering manoeuvres**
4. **Vary the speed**
5. **Drive straight**
6. **Remain stationary for 2–3 minutes**, logging static data
7. **Close the mission** in TMI
8. **Wait for the Control Unit power button light to go out — up to 90 seconds**[TRIMBLE METHOD] MX60 QSG Rev B, §5.5 p.13 — Trimble's documented sequence. It instructs

that the mission be finalized *"according to the following sequence"*, and gives the reason rather than an obligation: symmetrical start and end *"supports forward and reverse processing modes in the office software"*. **Not a manufacturer requirement — and still the right thing to do**, because above is a fact about the data regardless of who requires what. Whether Parametrix makes it mandatory is **SOP**.

21.2 It is initialization, backwards

Steps 3–6 are the initialization sequence in reverse order MX60 QSG Rev B, p.13 — manoeuvres, speed variation, straight, static, where the start was static, straight, speed variation, manoeuvres.

That is not a mnemonic. The backward pass runs through the data in reverse, so what it meets first is what the forward pass met last.

21.3 The mistake

IMPORTANT

Navigation logging stops the instant the mission is closed. Closing the mission and *then* driving to open sky achieves nothing. The manoeuvres have to be inside the logged data.

21.4 Five minutes

It takes about five minutes and it is the cheapest quality improvement available in the entire workflow.

You cannot append it the next day — a new mission is a new trajectory. There is no office procedure that buys it back (*Technical Manual §14.3*).

21.5 Record

That the closing sequence was performed, and the time. The office will look for it.

IN PLAIN LANGUAGE

What this section means. The trajectory is computed twice — forwards through the mission and backwards through it — and the two are merged. The closing sequence is what the backward pass starts from.

Why it matters. A bad stretch in the middle of a mission has good data on both sides of it, so the merge bridges it. A bad stretch at the *end* has good data on one side only. The closing sequence is what gives the end of the mission its second side.

Remember this. Drive to open sky **with the mission still running**, do the manoeuvres, then sit still for two to three minutes before closing. Logging stops the instant the mission closes.

If this is skipped. The last part of the corridor — often the part you were most pressed for time on — is the weakest part of the dataset, and it cannot be repaired the next day, because a new mission is a new trajectory.

22 Shutdown

22.1 Do

1. Confirm the mission is **closed**
2. **Wait for the Control Unit power button light to go out** — up to 90 seconds
3. Only then remove power or the data disk

(MX60 UG Rev B)

22.2 Stop if

CAUTION

Do not remove power or the data disk while the power button light is on.
The mission may not be fully written.

22.3 Then go to §23

Before the vehicle moves. Everything in §23 is recoverable here in minutes and costs a mobilisation from the office.

23 Field Close-out

Before the vehicle leaves the corridor. Not at the yard.

23.1 Coverage — against the mission plan

☐	CHECK
☐	Every planned pass driven, in the planned direction
☐	The corridor driven end to end, including any extent beyond the deliverable needed to bracket control
☐	Planned overlap actually collected — this is the one that removes office options if missed
☐	Sections not collected, and why, recorded
☐	Closing sequence performed (§21)

23.2 Data — on the disk

☐	CHECK
☐	Mission folder present, with a plausible size
☐	Run count matches what was driven
☐	<code>POS_1/raw</code> present and not empty
☐	Base station data captured, if a local base was used
☐	Power button light out before the disk was disturbed

`POS_1/raw` **is the one that cannot be recovered.** Without it there is no post-processed trajectory and the mission is NAV-only (*Technical Manual §5.2*).

23.3 Record

☐	CHECK
☐	Field record complete (§26, Appendix C)
☐	Comments recorded during collection (§19)

☐	CHECK
☐	Any deviation from the plan written down, with its reason

23.4 Then decide

☐	CHECK
☐	Any re-drive decided and performed now (§24)

CAUTION

While you are still at the site you can re-drive a run. Ten minutes down the road, you cannot.

24 Re-collect or Not

[RECOMMENDED] — everything in this section is a recommendation from this project. **Who decides a re-collection is and, and neither is settled** (§24.5). What is not proposed is the arithmetic: on site a re-drive costs twenty minutes, and from the office it costs a mobilisation.

24.1 The rule of thumb

On site, a re-drive costs twenty minutes. From the office, it costs a mobilisation.

So the bar for re-driving **while you are still here** is much lower than it feels.

24.2 Re-drive now

A run ended unexpectedly — power cut, disk, sensor fault	It has no closing sequence, and may be incomplete
A sensor was down for part of a run	You will not know what is missing until the office opens it
A planned pass was not driven	
Planned overlap was not collected	Removes LiDAR QC and run-to-run for that stretch
Navigation status was orange or red for a significant stretch	§14.4
Occlusion spoiled a stretch you cannot re-drive later	

24.3 Cannot be fixed by re-driving

A mission with no closing sequence	A new mission is a new trajectory. Re-drive the whole mission, or accept it
Weak initialization	The same — it is per-mission
A GNSS outage longer than the published envelope	Not a driving problem. Record it and raise it (§17.4)

24.4 If you decide not to re-drive

Record the decision and the reason. The office needs to know it was a decision and not an oversight.

24.5 If the segment cannot be served at all

Where no amount of re-driving will produce an acceptable result, the honest finding is that **mobile mapping may not be the appropriate acquisition method for that segment**. Record it and raise it — that is a project decision, not yours to absorb.

SET BY THE PROJECT — WHO DECIDES, AGAINST WHAT, AND WHAT THE CLIENT IS TOLD

(SOP §22.5).

IN PLAIN LANGUAGE

What this section means. Deciding, while you are still on site, whether what you have is good enough — because the cost of that decision changes enormously once you leave.

Why it matters. On site, a re-drive is twenty minutes. From the office it is a mobilisation: crew, vehicle, traffic control, and a corridor that may have changed.

Remember this. Check coverage and the disk before the vehicle moves off. If you are unsure whether a pass was good, drive it again — that is the cheapest decision available to you all day.

If this is skipped. The gap is found in the office, and somebody has to choose between going back and delivering something with a hole in it.

25 Data Transfer and Handoff

25.1 Removing the disk

1. Confirm the mission is closed and the **power button light is out** (§22)
2. Remove the exchangeable data disk
3. **Never connect the USB cable while the disk is inside the Control Unit**

MX60 UG Rev B, p.10

25.2 What goes to the office

The complete mission folder — not the `.mxdb` alone.

```
TMX<serial>-<mission id>/
├─ Base/ base RINEX, if collected
├─ Camera_1 ... Camera_4/ imagery
├─ Laser_1, Laser_2/ raw scanner data
├─ POS_1/raw/ ← the one that cannot be recovered
├─ Extcal.json the calibration file
└─ <mission>.mxdb an index, not the data
```

The `.mxdb` is a table of contents. Copying it alone copies nothing useful (*Technical Manual §5.2*).

Plus: the **field record** (§26), base station data if a local base was occupied, and any deviation from the plan with its reason.

25.3 The disk rule

CAUTION

Do not clear a data disk until a verified copy exists in at least two locations and the `.mxdb` has been confirmed to open.

A cleared disk is not recoverable, and a mobile mapping mission is not re-drivable at reasonable cost.

[SOP §11.2] — the *confirmation-in-writing* step is a Parametrix practice.

[EQUIPMENT] The underlying fact is not proposed. A cleared disk is not recoverable and a mission is not re-drivable at reasonable cost. **Not on a promise, not on a message, not because the disk is needed tomorrow.**

25.4 Handoff is a transfer of responsibility

Until the office has confirmed that verified copy, **the field holds the only copy of an unrepeatable measurement.** Close-out is not complete when the vehicle is packed; it is complete when the office confirms.

25.5 Record

What was transferred, to whom, when.

SET BY THE PROJECT — THE TRANSFER AND VERIFICATION PROCEDURE,
the folder structure and naming, and whether a chain-of-custody record is required (*SOP §11*).

26

What Must Accompany the Data

26.1 The field record

[SOP §9.6] — what the record contains is not yet decided.

No software produces it. That part is a fact, not a proposal: it is the only record of everything the software cannot see, and the office depends on it.

Recorded **at the time**, per mission:

Date, operator, vehicle, mission ID	
Capture settings used — and which presentation you saw (§10.2)	
Initialization location and time	
Time green was reached, and time significant recording began	The settling time (§14.5)
Each run — start, end, any incident	
GNSS conditions observed , per hostile stretch	Entry/exit times, whether overlap was driven
Weather	
Traffic and occlusion events	
Anything not collected, and why	
Closing sequence performed — and the time	
Disk and free space at end	
Any re-drive decision, and its reason	§24

Template: **Appendix C**.

26.2 Why each of the hard ones matters

Conditions and occlusion	Nothing in the software knows a truck was there

What was not collected, and why	Otherwise the office cannot tell a gap from a decision
The settling time	It is the only evidence §14.2 was respected
Closing sequence performed	The office looks for it, and cannot tell from the data alone

26.3 With the data

☐	
☐	The complete mission folder (§25.2)
☐	The field record
☐	Base station data, if a local base was occupied
☐	Deviations from the mission plan, with reasons

26.4 The test

Could somebody who was not there work out what happened on this corridor, three weeks from now, from what you are handing over?

If not, the missing piece is in the field record.

27 Common Problems

27.1 Troubleshooting pattern

When something is wrong, do not start changing settings at random.

1. **Name what you actually see.** A color, missing device, alarm, status or screen message is more useful than "the MX60 is not working."
2. **Return to the last known good state.** Check power, connections, TMI connection, Vehicle Settings and the current mission before changing anything else.
3. **Change one thing at a time.** If three things are changed together, you do not know what fixed the problem or what new problem was introduced.
4. **Record the abnormal condition.** Use a TMI comment where appropriate and put the event in the field record.
5. **Stop when data integrity is uncertain.** A run that looks complete but was collected with a missing sensor, failed initialization or interrupted closing sequence is not made safer by continuing to drive.

27.2 Symptom table

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP, RE-DRIVE OR RAISE WHEN
TMI will not load	Confirm the Control Unit is fully started and the field device is on its network	Enter <code>http://tmi.mx-scan.net</code> exactly, use Chrome, then repeat §9 from the beginning	You cannot establish a stable TMI connection; do not invent an IP/SSID or change system networking from memory
System will not start	Power button held 15 s ? Supply live? Battery healthy?	§6, then §8	Power or Control Unit status is uncertain
LEDs blink and do not settle	Allow for a firmware update, which can take up to 6 minutes	Compare the LED condition with §8.2	The condition persists beyond the documented startup behavior
A sensor is missing from the device list	Compare the list with the confirmed system configuration, then check cable and power to that sensor	Restart only by the documented sequence in §8 and re-check §9.3	Stop. Do not collect with a required sensor missing

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP, RE-DRIVE OR RAISE WHEN
A fitted DMI or GAMS logs nothing	Is it activated in Vehicle Settings?	Verify the installed configuration against §10.1	The sensor is expected for the mission and still does not log
Navigation will not reach green	Ask TMI which parameter is holding it	Heading: drive more dynamic manoeuvres. Position: move to better sky view	Initialization does not converge before meaningful collection
Status is stuck on heading	Dynamic manoeuvres and whether GAMS is available	More turns and speed changes per §14.3	Heading quality does not recover
Status is stuck on position	Sky view	Move the vehicle. Waiting in the same blocked location will not help	Position quality does not recover
Status goes orange mid-corridor	Identify the location and likely obstruction	Record it, maintain planned overlap and use §17	The degraded section cannot be covered defensibly or the guide calls for re-drive
Audible alarm	Battery Protect	Restore charge immediately, then follow §7	Power stability is not restored
Power cut mid-run	Treat the run as ended	Record the event and use §24	The run has no valid closing sequence. Make the re-drive decision now
Storage is filling faster than expected	Remaining free space and expected run duration	End the run cleanly before the disk fills if needed	Continuing risks an interrupted run or missing closing sequence
Capture settings look different from the guide	Confirm TMI version and which presentation is shown	Record what you see and compare §10.2	A required setting cannot be positively identified
Cannot find 2000 kHz on screen	Remember 2000 kHz is the system total	TMI uses per-scanner 500/1000 kHz values; see §10.2	The actual configured rate still cannot be verified
Imagery looks wrong on screen	Rain, dust, insects or smear on optics	Clean the optics and inspect again	The affected stretch was collected with unusable imagery and imagery is required

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP, RE-DRIVE OR RAISE WHEN
A planned pass was skipped	Record exactly which pass and why	Use §24 before leaving	Missing overlap or coverage removes an office remedy
A run was stopped accidentally but the mission is still open	Confirm the mission itself remains open	Record the interruption, leave the mission open, and start a new run when ready (§20)	Coverage or overlap was lost; use §24 before leaving
Mission was closed before the closing sequence	A closed mission cannot be appended to	Record it; any new mission needs its own initialization and closing sequence (§20.4)	Use §24 — a missing mission closing sequence is not repaired by starting another mission
Transfer copy looks incomplete	Compare file count and total size with the source	Confirm the <code>.mxd</code> opens and <code>POS_1/raw/</code> is present and non-empty; use SOP §11.2	Do not clear the field disk until verification is complete
Unsure whether to re-drive	Use the actual missing or abnormal condition, not the schedule	§24	If you cannot explain why the existing data is adequate, raise it before leaving site

27.3 When the answer is "ask"

Stop and ask rather than improvising on the vehicle. You may always stand down, record a comment, or re-drive while you are still on site (§1.4).

Escalation is the correct result when the guide reaches the end of its troubleshooting path. The guide is supposed to help a minimally experienced MX60 operator recognize that point; it is not supposed to make them invent a new field procedure.

A Preflight Checklist

Printable. One per mission.

Date _____ Operator _____ Vehicle _____

A1 • Before leaving the yard

☐	\$
☐ Mission plan read — route, passes and directions, overlap, hostile stretches	2
☐ Two initialization locations identified	2
☐ Calibration currency confirmed	2
☐ Field record form and this guide on board	2

A2 • Equipment

☐	\$
☐ Optics clean — scanner windows, camera dome	3
☐ Cables sound, no chafe or damage	3
☐ Rack and fixings secure, per the rack's own guide	3
☐ Sensor Unit seated as measured — two people to mount	3, 4
☐ GAMS secure and same antenna type, if fitted	13
☐ DMI secure on a non-steering wheel, if fitted	3
☐ Cables routed clear of doors and moving parts	5

A3 • Power

☐	\$
☐ Supply rated 30 A or more , fused 35 A near the battery	6
☐ Battery healthy and charged	6

☐		\$
☐	Auxiliary battery healthy, if fitted	6
☐	No Battery Protect alarm during preflight	7

A4 · Start-up

☐		\$
☐	Ignition on, supply live	8
☐	Power button held ≥ 15 s	8
☐	LEDs settled to ready	8
☐	TMI reached in Chrome at <code>tmi.mx-scan.net</code>	9
☐	Every sensor present in the device list	9.3

A5 · Settings

☐		\$
☐	DMI activated in Vehicle Settings , lever arm and side set — if fitted	10.1
☐	GAMS activated in Vehicle Settings , lever arm set — if fitted	10.1
☐	Install Height preset correct for this vehicle	10.1
☐	Capture settings set — and which presentation you saw recorded	10.2
☐	Dust filter set deliberately	10.2

A6 · Disk

☐		\$
☐	Correct disk, seated	11
☐	Free space sufficient for the planned mission, with margin	11
☐	USB cable not connected with the disk inside	11

A7 · Initialization

☐		\$
☐	Parked in genuinely open sky	12
☐	Mission started	12
☐	2–3 minutes static	12
☐	Straight ~20 m	12
☐	Speed varied and dynamic steering performed	12
☐	Status reached green — time recorded	14.5
☐	Settling time allowed — up to 10 minutes	14.2
☐	Time significant recording began — recorded	14.5

Signed _____

B End-of-Mission Checklist

Printable. Complete **before the vehicle leaves the corridor**.

Date _____ Mission ID _____ Operator _____

B1 · The closing sequence

☐		§
☐	Last run finished	21
☐	Drove to open sky with the mission still running	21.3
☐	Dynamic steering manoeuvres	21
☐	Speed varied	21
☐	Drove straight	21
☐	Stationary 2–3 minutes, logging	21
☐	Mission closed in TMI — time recorded	21
☐	Power button light out — up to 90 s — before power or disk disturbed	22

It cannot be added later. If it was missed, record that plainly and go to §24.

B2 · Coverage — against the mission plan

☐		§
☐	Every planned pass driven, in the planned direction	23.1
☐	Corridor driven end to end, including extent needed to bracket control	23.1
☐	Planned overlap actually collected	17.3
☐	Sections not collected, and why, recorded	23.1

B3 • Data on the disk

☐	§
☐ Mission folder present, plausible size	23.2
☐ Run count matches what was driven	23.2
☐ POS_1/raw present and not empty	23.2
☐ Base station data captured, if a local base was used	23.2

B4 • Record

☐	§
☐ Field record complete (Appendix C)	26
☐ Comments recorded during collection	19
☐ Deviations from the plan written down, with reasons	23.3
☐ Time green reached, and time significant recording began	14.5

B5 • Decide

☐	§
☐ Any re-drive decided and performed now	24
☐ If not re-driving, the decision and reason recorded	24.4

CAUTION

While you are still at the site you can re-drive a run. Ten minutes down the road, you cannot.

B6 • Handoff

☐	§
☐ Complete mission folder transferred — not the .mxd alone	25.2

☐		\$
☐	Field record transferred	26
☐	Disk NOT cleared — not until the office confirms a verified copy	25.3

Signed _____

C Field Record Form

No software produces this. It is the only record of what the office cannot see.

C1 · Mission

- Date
- Operator
- Vehicle
- **Mission ID**
- Project
- Weather at start
- Weather changes during

C2 · Configuration

Capture settings used	
Which presentation was shown	☐ Measurement Prog + Line Speed ☐ Laser Mode
Lateral Range Limit	
Dust filter	☐ off ☐ on
GAMS fitted and activated	☐ n/a ☐ fitted, activated ☐ fitted, NOT activated
DMI fitted and activated	☐ n/a ☐ fitted, activated ☐ fitted, NOT activated · side: ☐ left ☐ right

C3 · Initialization

Location	
Time started	
Static period	_____ min
Straight run performed	☐
Speed variation and manoeuvres performed	☐

Time green reached
Time significant recording began
Settling time allowed _____ min

C4 · Runs

RUN	START	END	INCIDENTS

C5 · GNSS-hostile stretches

Planned overlap driven?

- ☐ yes ☐ no
- ☐ yes ☐ no

C6 · Events

TIME	EVENT	EFFECT ON THE DATA

Traffic, occlusion, stops, a lane you could not use, a turnaround inside the corridor, a Battery Protect alarm, a sensor dropout.

C7 · Not collected

Re-drive?

- ☐ done ☐ not — reason:

C8 · Close-out

Closing sequence performed	☐ yes — time: _____ ☐ no — reason:
Mission closed at	
Power light out before disk removed	☐
Run count on disk	_____ · matches driven: ☐
POS_1/raw present	☐
Base station data captured	☐ n/a ☐ yes
Free space at end	

C9 · Handoff

Transferred to	
Date and time	
Disk cleared?	☐ NO — awaiting office confirmation ☐ yes, confirmation received from:

Signed _____

D TMI Status and Warning Reference

NOT DOCUMENTED BY TRIMBLE

The TMI version installed on the Parametrix system has not been confirmed, and TMI's interface has changed between versions. This appendix is sourced to TMI User Guide Rev L and MX60 Quick Start Guide Rev B. Verify against the system before issue.

D1 · Reaching TMI

Browser	Chrome
Capture	<code>http://tmi.mx-scan.net</code>
Administration	<code>http://admin.mx-scan.net</code>
Modules	TMI.Capture · TMI.AI

(TMI UG Rev L)

Served by the Control Unit. No internet access required; the addresses resolve only on the Control Unit's network.

D2 · Status colours

COLOUR	MEANING
Green	The parameter meets its accuracy threshold
Orange	Degraded but operating. Recording is permitted
Red	Not meeting threshold

CAUTION

Orange permits recording. Survey-grade work does not. Until / is decided, treat orange as a stop-and-assess condition and record it.

Green means the solution met its accuracy figures — not that it has finished converging. Trimble asks for **up to ten more minutes** of settling before recording data that matters (MX60 QSG Rev B).

D3 · When the navigation status will not reach its ready state

HOLDING PARAMETER	LIKELY CAUSE	ACTION
Heading	Insufficient dynamic manoeuvres, or GAMS unavailable	More turns and speed changes. Heading is the hardest component to resolve
Position	Poor sky view	Move to a genuinely open location
Attitude	Insufficient motion variety	Complete the full manoeuvre profile

Ask TMI which parameter is holding the solution rather than waiting.

Heading means drive more; position means move the vehicle. **Waiting helps neither.**

D4 · Alarms and protective behaviour

INDICATION	MEANING	RESPONSE
Audible alarm	Battery Protect — supply below 10.5 V for more than 12 s	Power cuts at 90 s , leaving roughly 78 s . Recovery needs the voltage above 12.0 V within that window. Restore charge immediately; do not continue MX60 UG Rev B, p.27
Sensor absent from the device list	Cable, power or sensor fault	Stop. A run with a sensor down is incomplete (§24)
Storage warning	Disk filling	Reassess before it fills mid-run — an interrupted run loses the closing sequence (§21)

D5 • Capture settings

Two presentations exist, depending on the TMI version (CONFLICT-005):

Source | Presentation

---|---

MX60 QSG Rev B, p.10 | **Measurement Prog** [500 kHz, 1000 kHz] and
Line Speed [120 Hz, 200 Hz]

TMI UG Rev L, p.29 | A single combined **Laser Mode**

Expect either. Record which you saw.

- The measurements-per-second values shown in the interface are **rounded**
TMI UG Rev L, pp.28–29
- **Lateral Range Limit** is settable **5–50 m** TMI UG Rev L, p.29. Whether it affects accuracy or is purely a data-volume control is
- A **dust filter** is available, intended for **unpaved roads and mine sites** (*Product Bulletin, January 2025*)

A resolved conflict, recorded so it is not re-opened. The specification sheet quotes **1000 / 2000 kHz** and **240 / 400 Hz** where the User Guide quotes **500 / 1000 kHz** and **120 / 200 Hz** — a factor of exactly two, being system totals against per-scanner figures. **TMI uses the per-scanner numbering.**

RESOLVED-001 .

D6 • Comments

Recorded during collection, against the moment. §19.

D7 • Calibration import

TMI accepts a boresight calibration JSON via **USB1** TMI UG Rev L, p.18. Relevant where a calibration is computed in the office and applied to the system rather than to a project.

D8 • Administration

Firmware update	Copy the <code>.tmx.install</code> file to a USB stick, connect to USB1 , Refresh then Install . The system shuts down; wait until all Control Unit LEDs are off, then power back on. Powering on may take up to six minutes TMI UG Rev L, p.49
Licence update	Copy the licence file to a USB stick, connect to USB1, Import TMI UG Rev L, p.53
Remote access	When remote control is given to Trimble Support, all data on the system is visible to them. Remove confidential data before granting it TMI UG Rev L, p.54
Wi-Fi country	Must be set on first use and every time you collect in a different country MX60 UG Rev B, p.48
Wi-Fi limitation	Cannot connect to a hotspot that requires logging in on a web page TMI UG Rev L, p.51

This appendix is a reference for the indications, not a substitute for the procedure. The procedure is §8 to §22.

E Quick Card

The ten things that cost the most if missed. One sheet, both sides. Keep it in the vehicle.

#		WHY IT COSTS
1	Activate DMI and GAMS in Vehicle Settings — installed and wired is not enough	A sensor that is not activated logs nothing , and nothing looks wrong until the office opens it (§10.1)
2	Park in genuinely open sky to start	The static period is doing real work. Under trees, half of it is wasted (§12)
3	Do the full manoeuvre profile — straight 20 m, vary speed, turn	Heading is the hardest thing the system has to solve, and turning is how it solves it (§12.2)
4	Green is not finished. Allow up to 10 more minutes before recording anything that matters	The first data after green is the weakest data of the day. Nothing tells you when it improves (§14.2)
5	Drive every planned pass, in the planned direction	Direction is not cosmetic. A reverse pass fixes the far side of the corridor (§15.3)
6	Drive the planned overlap	Missing it removes LiDAR QC and run-to-run registration for that stretch. No warning, no way to tell afterwards (§17.3)
7	Use Comments as things happen	Nothing in the software knows a truck was there. Ten seconds now, an hour of office guesswork later (§19)
8	Do the closing sequence — before closing the mission	Logging stops the instant the mission closes. It cannot be added later. Five minutes (§21)
9	Check coverage and the disk before the vehicle moves	On site a re-drive is twenty minutes. From the office it is a mobilisation (§23)
10	Do not clear a disk until the office confirms a verified copy	A cleared disk is not recoverable, and the mission is not re-drivable at reasonable cost (§25.3)

And one alarm

CAUTION

Audible alarm = Battery Protect. Supply has been below **10.5 V for 12 seconds**. Power cuts at **90 seconds**, so you have roughly **78 seconds** to restore charge — the voltage has to rise above **12.0 V** within that window.

Restore charge immediately. An interrupted run loses the closing sequence with it.

And one thing you may always do

Stand down. On safety or data-quality grounds, without asking first. Record the decision and the reason. **It is not conditional on you being right** (*SOP §9.5*).



Parametrix MX60 Field How To